Lecture 6: Generalized Linear Models

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1. Review and Continuation: Logistic Regression

2. Exponential Families and Generalized Linear Models
   - Motivation
   - Exponential family
   - From Parameter Estimation to GLMs
   - Optional reading: extensions and proofs

Reading materials & Reference

Reading materials:

- Chapter 3, Stanford CS 229 Lecture Notes, `https://cs229.stanford.edu/notes2022fall/main_notes.pdf`
- Lecture 4, Stanford CS 231n, `http://cs231n.stanford.edu/schedule.html`

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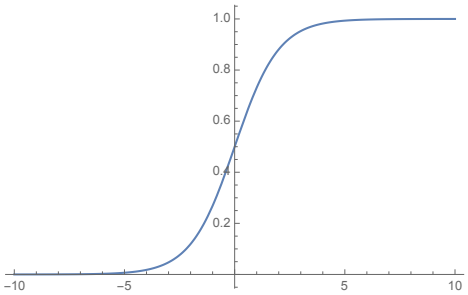
  But we can use the data to learn the distribution (by assuming the data distribution).

The logistic function

Consider first of all the case of two classes.
The posterior probability for class $\mathcal{C}_1$:

$$p(\mathcal{C}_1|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \quad (2)$$

$$= \frac{1}{1+\exp(-z)} = \sigma(z) \quad (3)$$



Properties of the logistic function:

- $1 - \sigma(z) = \sigma(-z)$
- $\sigma'(z) = \sigma(z)\,(1 - \sigma(z))$

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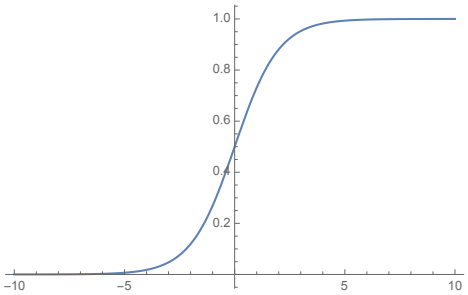
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where we have defined

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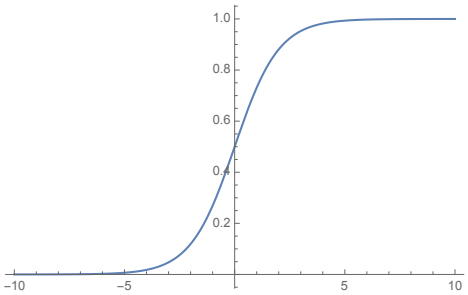
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For $K > 2$, define $z_k = \log[p(\mathbf{x} \mid \mathcal{C}_k)p(\mathcal{C}_k)]$. Then

$$p(\mathcal{C}_k|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_j p(\mathbf{x}|\mathcal{C}_j)p(\mathcal{C}_j)} = \frac{\exp(z_k)}{\sum_j \exp(z_j)} \quad (5)$$



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Logistic Regression: fix the notation

Let $\mathbf{x} \in \mathbb{R}^d$ be an input, $\mathbf{w} \in \mathbb{R}^d$ the weights, and $b \in \mathbb{R}$ the intercept. The label is $Y \in \{0, 1\}$.

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\begin{aligned}
z(\mathbf{x}) &= \mathbf{x}^\top \mathbf{w} + b, \\
p(1 \mid \mathbf{x}; \mathbf{w}, b) &= \sigma(z(\mathbf{x})), \\
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- $\sigma(\cdot)$ is the sigmoid; z is a real-valued score, not a probability.
- Training observations are $(\mathbf{x}_n, y_n)$. The design matrix $\mathbf{X}$ has rows $\mathbf{x}_n^\top$; $\mathbf{y} = (y_1, \ldots, y_N)^\top$.

MLE estimates the model parameters

Treat the observed inputs as fixed and assume independent labels given the inputs. The conditional likelihood is

$$L(\mathbf{w}, b) = p(\mathbf{y} \mid \mathbf{X}; \mathbf{w}, b) = \prod_{n=1}^{N} p(y_n \mid \mathbf{x}_n; \mathbf{w}, b).$$

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L denotes likelihood; $J = -\log L/N$ denotes average negative log-likelihood. The inputs after $|$ are conditioned on; quantities after $;$ are model parameters.

Review: from likelihood to the objective

For $y_n \in \{0, 1\}$, let $z_n = \mathbf{x}_n^\top \mathbf{w} + b$ and $p_n = \sigma(z_n)$. The intercept remains explicit during this review.

$$L(\mathbf{w}, b) = \prod_{n=1}^{N} p_n^{y_n} (1 - p_n)^{1 - y_n},$$

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- Maximizing L is equivalent to minimizing the average negative log-likelihood J.
- ℓ is the loss for one observation; $J = \frac{1}{N} \sum_n \ell(z_n, y_n)$ averages it.

Review / continuation: From the gradient to an update

For one example, $\ell(z, y) = \log(1 + e^z) - yz$, $z = \mathbf{x}^\top \mathbf{w} + b$, $p = \sigma(z)$.

$$\frac{\partial \ell}{\partial z} = \sigma(z) - y = p - y,$$

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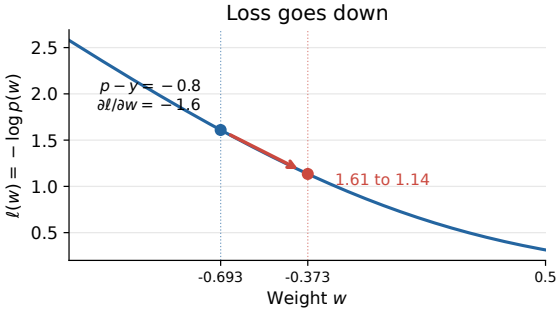
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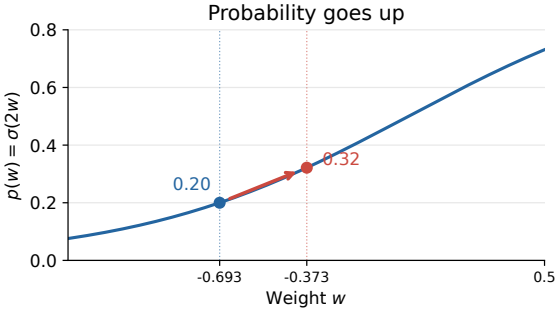
$$\nabla_{\mathbf{w}} \ell = (p - y)\mathbf{x}, \qquad \frac{\partial \ell}{\partial b} = p - y.$$

Review / continuation: From the gradient to an update

See the update: $x = 2$, $y = 1$, $p = 0.2$, $b = 0$ fixed; update only w with $\alpha = 0.2$.

$$\frac{\partial \ell}{\partial w} = -1.6 \quad \Longrightarrow \quad w_{\text{new}} = w - 0.2(-1.6) = w + 0.32.$$





Review / continuation: From one example to a training algorithm

Let $\mathbf{X} \in \mathbb{R}^{N \times d}$ have rows $\mathbf{x}_n^\top$ and $\boldsymbol{p} = \sigma(\mathbf{X}\mathbf{w} + b\mathbf{1})$ elementwise.

$$\nabla_{\mathbf{w}} J = \frac{1}{N} \sum_{n=1}^{N} (p_n - y_n)\mathbf{x}_n \qquad \text{(average the sample gradients)}$$

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3. Update with learning rate $\alpha > 0$: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}} J$, $b \leftarrow b - \alpha\, \partial J / \partial b$.

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The stationary equation has no general closed-form solution; repeat these steps.

Review / continuation: One gradient step by hand

Given $x = 2$, $y = 1$, $w = b = 0$, and learning rate $\alpha = 0.1$:

Try first: should the probability of class 1 increase or decrease?

Review / continuation: One gradient step by hand

Given $x = 2$, $y = 1$, $w = b = 0$, and learning rate $\alpha = 0.1$:

1. **Predict:** $z = wx + b = 0$, so $p = \sigma(0) = 0.5$.

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2. **Compute the residual:** $r = p - y = 0.5 - 1 = -0.5$.
   Hence $\partial \ell / \partial w = rx = -1$ and $\partial \ell / \partial b = r = -0.5$.

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3. **Subtract the gradient:** $w_{\text{new}} = 0 - 0.1(-1) = 0.1$, $b_{\text{new}} = 0 - 0.1(-0.5) = 0.05$.
4. **Predict again:** $z_{\text{new}} = 0.1 \cdot 2 + 0.05 = 0.25$, so $p_{\text{new}} \approx 0.562$.

Check: $-\log 0.5 = 0.693$ decreases to $-\log 0.562 \approx 0.576$.

Review / continuation: Regularized logistic regression: the same workflow returns

A gradient step improves training fit; we also want to control coefficient size.

1. Add a penalty: weights are penalized; the intercept is not.

$$J_\lambda(\mathbf{w}, b) = J(\mathbf{w}, b) + \frac{\lambda}{2}\|\mathbf{w}\|_2^2, \qquad \lambda > 0.$$

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2. Differentiate the new term: $\nabla_{\mathbf{w}}(\frac{\lambda}{2}\sum_j w_j^2) = \lambda \mathbf{w}$.

$$\nabla_{\mathbf{w}} J_\lambda = \frac{1}{N}\mathbf{X}^\top(\boldsymbol{p} - \mathbf{y}) + \lambda \mathbf{w}.$$

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- Fit feature scaling on training data; reuse it for validation and test data.
- Fit $(\mathbf{w}, b)$ on training data for each candidate λ.

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A gradient step improves training fit; we also want to control coefficient size.

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$$J_\lambda(\mathbf{w}, b) = J(\mathbf{w}, b) + \frac{\lambda}{2}\|\mathbf{w}\|_2^2, \qquad \lambda > 0.$$

2. Differentiate the new term: $\nabla_{\mathbf{w}}(\frac{\lambda}{2}\sum_j w_j^2) = \lambda\mathbf{w}$.

$$\nabla_{\mathbf{w}} J_\lambda = \frac{1}{N}\mathbf{X}^\top(\boldsymbol{p} - \mathbf{y}) + \lambda\mathbf{w}.$$

- Fit feature scaling on training data; reuse it for validation and test data.
- Fit $(\mathbf{w}, b)$ on training data for each candidate λ.
- Select λ on validation data; evaluate once on untouched test data.

Review / continuation: Regularized logistic regression: the same workflow returns

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- Fit $(\mathbf{w}, b)$ on training data for each candidate λ.
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Stronger shrinkage can reduce sensitivity, but excessive shrinkage can underfit.

Quick check: which regularization strength?

Two models use the same data and features. Which λ would you choose?

A. $\lambda = 0.01$: training NLL $= 0.20$, validation NLL $= 0.70$.

B. $\lambda = 1$: training NLL $= 0.35$, validation NLL $= 0.45$.

C. Always choose the model with the smaller training NLL.

Think / vote: choose an answer and give one short reason.

Quick check: which regularization strength?

Two models use the same data and features. Which λ would you choose?

A. $\lambda = 0.01$: training NLL $= 0.20$, validation NLL $= 0.70$.

B. $\lambda = 1$: training NLL $= 0.35$, validation NLL $= 0.45$.

C. Always choose the model with the smaller training NLL.

Answer: B.

Choose $\lambda = 1$ because its validation NLL is smaller.
Compare the same unpenalized loss; keep the test set for the final evaluation.

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.

Least squares

Known prediction rule: $\hat{y} = z$.

The output is a real-valued prediction of Y.

$$\hat{y}(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}.$$

This is the linear predictor from our regression lectures.

Logistic regression

Known probability: $\Pr(Y = 1 \mid \mathbf{x}) = \sigma(z)$.

The output is the probability of class 1.

$$\Pr(Y = 1 \mid \mathbf{x}) = \sigma(\mathbf{x}^\top \mathbf{w}).$$

This is the probability predictor from our classification lectures.

We already know these prediction rules. Next, relate both to $\mathbb{E}[Y \mid \mathbf{x}]$.

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.

Least squares

Known prediction rule: $\hat{y} = z$.

Recall the additive-noise model

$$Y = z + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid \mathbf{x}] = 0.$$

Therefore its conditional mean is

$$\begin{aligned} \mu := \mathbb{E}[Y \mid \mathbf{x}] &= z + \mathbb{E}[\varepsilon \mid \mathbf{x}] \\ &= z. \end{aligned}$$

Logistic regression

Known probability: $\Pr(Y = 1 \mid \mathbf{x}) = \sigma(z)$.

Use the two possible labels

$$Y \in \{0, 1\}, \qquad \Pr(Y = 1 \mid \mathbf{x}) = \sigma(z).$$

Therefore its conditional mean is

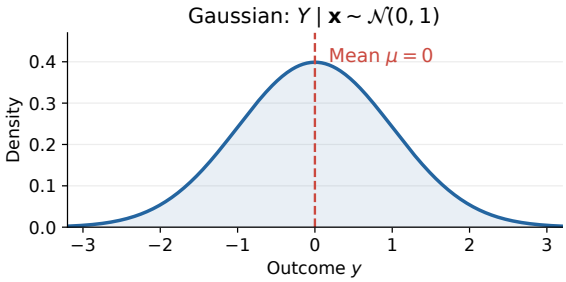
$$\begin{aligned} \mu := \mathbb{E}[Y \mid \mathbf{x}] &= 0 \cdot [1 - \sigma(z)] + 1 \cdot \sigma(z) \\ &= \sigma(z). \end{aligned}$$

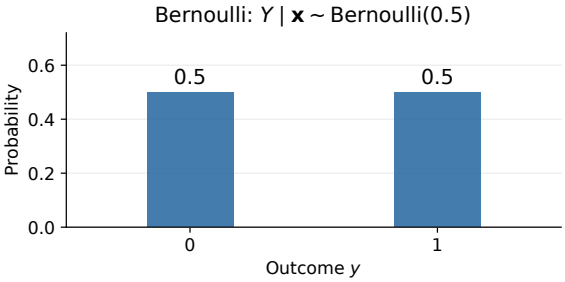
Both model $\mu = \mathbb{E}[Y \mid \mathbf{x}]$: least squares uses $\mu = z$; logistic uses $\mu = \sigma(z)$.

For binary labels, μ is the class-1 probability (the earlier p_n for sample n).

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.





Least squares: $\mu = z = 0$.

Logistic regression: $\mu = 0.5$.

How does each probability model give a training loss?

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.

Gaussian $\longrightarrow$ least squares

Mean: $\mu = z$, fixed $\sigma^2 > 0$.

Observation likelihood

$$p(y \mid \mathbf{x}; \mathbf{w}) = e^{-(y-z)^2/(2\sigma^2)} / \sqrt{2\pi\sigma^2}.$$

The observed value is scored by a Gaussian density.

Bernoulli $\longrightarrow$ logistic regression

Mean: $\mu = \sigma(z)$, $y \in \{0, 1\}$.

Observation likelihood

$$p(y \mid \mathbf{x}; \mathbf{w}) = \mu^y (1-\mu)^{1-y}.$$

The observed label is scored by its Bernoulli probability.

Same fitting principle: choose parameters that make the observed data likely.

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.

Gaussian $\longrightarrow$ least squares

Mean: $\mu = z$, fixed $\sigma^2 > 0$.

Negative log-likelihood

$$\ell_G(z, y) = \frac{(y - z)^2}{2\sigma^2} + C.$$

$C = \frac{1}{2}\log(2\pi\sigma^2)$ is independent of $\mathbf{w}$.

Bernoulli $\longrightarrow$ logistic regression

Mean: $\mu = \sigma(z)$, $y \in \{0, 1\}$.

Negative log-likelihood

$$\begin{aligned}\ell_B(z, y) &= -y \log \mu - (1 - y) \log(1 - \mu) \\ &= \log(1 + e^z) - yz.\end{aligned}$$

In both columns, $\ell = -\log p(y \mid \mathbf{x}; \mathbf{w})$.

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.

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Mean: $\mu = z$, fixed $\sigma^2 > 0$.

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Fit: squared error

$\sum_n (y_n - \mathbf{x}_n^\top \mathbf{w})^2$.

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Fit: cross-entropy

Use $\mu_n = \sigma(\mathbf{x}_n^\top \mathbf{w})$.

Independent observations: MLE minimizes $\sum_n \ell(z_n, y_n)$.

For Gaussian, dropping C and the positive fixed scale leaves least squares.

From familiar predictions to probability models

Use $z = \mathbf{x}^\top \mathbf{w}$; a constant feature absorbs the intercept.

Gaussian $\longrightarrow$ least squares

Mean: $\mu = z$, fixed $\sigma^2 > 0$.

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Use $\mu_n = \sigma(\mathbf{x}_n^\top \mathbf{w})$.

One fitting principle, two losses.

Can Gaussian and Bernoulli share one probability template?

Two losses — can we find one probability template?

Recall $\ell = -\log p$. For Gaussian, use $\sigma^2 = 1$ and $C = \frac{1}{2}\log(2\pi)$.

Gaussian

1. Negative log-likelihood

$$\ell_G = \frac{(y-z)^2}{2} + C.$$

Bernoulli

1. Negative log-likelihood

$$\ell_B = \log(1+e^z) - yz.$$

Step 1: start with the complete loss, including the Gaussian constant C.

Two losses — can we find one probability template?

Recall $\ell = -\log p$. For Gaussian, use $\sigma^2 = 1$ and $C = \frac{1}{2}\log(2\pi)$.

Gaussian

1. Negative log-likelihood

$$\ell_G = \frac{(y-z)^2}{2} + C.$$

2. Negate

$$-\ell_G = -\frac{(y-z)^2}{2} - C.$$

Bernoulli

1. Negative log-likelihood

$$\ell_B = \log(1+e^z) - yz.$$

2. Negate

$$-\ell_B = yz - \log(1+e^z).$$

Step 2: multiply both sides of $\ell = -\log p$ by -1: $\log p = -\ell$.

Two losses — can we find one probability template?

Recall $\ell = -\log p$. For Gaussian, use $\sigma^2 = 1$ and $C = \frac{1}{2}\log(2\pi)$.

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1. Negative log-likelihood

$$\ell_G = \frac{(y-z)^2}{2} + C.$$

2. Negate

$$-\ell_G = -\frac{(y-z)^2}{2} - C.$$

3. Exponentiate

$$p_G = \exp\left[-\frac{(y-z)^2}{2} - C\right].$$

Bernoulli

1. Negative log-likelihood

$$\ell_B = \log(1+e^z) - yz.$$

2. Negate

$$-\ell_B = yz - \log(1+e^z).$$

3. Exponentiate

$$p_B = \exp\left[yz - \log(1+e^z)\right].$$

Step 3: exponentiate both sides: $p = \exp(-\ell)$.

Here p_G and p_B denote the Gaussian density and Bernoulli probability at (z, y).

Two losses — can we find one probability template?

Recall $\ell = -\log p$. For Gaussian, use $\sigma^2 = 1$ and $C = \frac{1}{2}\log(2\pi)$.

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1. Negative log-likelihood

$$\ell_G = \frac{(y-z)^2}{2} + C.$$

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$$\ell_B = \log(1+e^z) - yz.$$

2. Negate

$$-\ell_B = yz - \log(1+e^z).$$

3. Exponentiate

$$p_B = \exp\left[yz - \log(1+e^z)\right].$$

Step 4: both expressions are exponentials. Now compare their terms.

Next: expand $(y-z)^2$ to find terms involving y, z, or both.

Gaussian: separate the data and parameter terms

For $Y \mid \mathbf{x} \sim \mathcal{N}(z, 1)$ with $z = \mathbf{x}^\top \mathbf{w}$,

$$p(y \mid \mathbf{x}; \mathbf{w}) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(y - z)^2\right]$$

Gaussian: separate the data and parameter terms

For $Y \mid \mathbf{x} \sim \mathcal{N}(z, 1)$ with $z = \mathbf{x}^\top \mathbf{w}$,

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\begin{aligned}
p(y \mid \mathbf{x}; \mathbf{w}) &= \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(y - z)^2\right] \\
&= \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{y^2}{2} + zy - \frac{z^2}{2}\right]
\end{aligned}
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&= \underbrace{\frac{e^{-y^2/2}}{\sqrt{2\pi}}}_{\text{data only}} \exp\left[\underbrace{zy}_{\text{parameter} \times \text{data}} - \underbrace{\frac{z^2}{2}}_{\text{parameter only}}\right].
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\end{aligned}
$$

The familiar Gaussian density has three pieces.

Next: rewrite Bernoulli and check whether the same structure appears.

Bernoulli: the same three pieces

Recall the binary mean: $\mu = \mathbb{E}[Y] = \Pr(Y = 1)$, with $0 < \mu < 1$.

$$p_B(y; \mu) = \mu^y (1 - \mu)^{1-y}, \qquad y \in \{0, 1\},$$

Step 1: write the probability of either binary outcome.

Bernoulli: the same three pieces

Recall the binary mean: $\mu = \mathbb{E}[Y] = \Pr(Y = 1)$, with $0 < \mu < 1$.

$$p_B(y; \mu) = \mu^y (1 - \mu)^{1-y}, \qquad y \in \{0, 1\},$$
$$\log p_B(y; \mu) = y \log \mu + (1 - y) \log(1 - \mu),$$

Step 2: take the logarithm; products become sums.

Bernoulli: the same three pieces

Recall the binary mean: $\mu = \mathbb{E}[Y] = \Pr(Y = 1)$, with $0 < \mu < 1$.

$$
\begin{aligned}
p_B(y;\mu) &= \mu^y (1-\mu)^{1-y}, \qquad y \in \{0,1\}, \\
\log p_B(y;\mu) &= y \log \mu + (1-y)\log(1-\mu), \\
&= \underbrace{\log \frac{\mu}{1-\mu}}_{\text{coefficient of } y}\, y + \log(1-\mu).
\end{aligned}
$$

Step 3: collect the terms multiplying y.

The coefficient is the familiar log-odds. Next, give this coefficient a name.

Bernoulli: the same three pieces

Recall the binary mean: $\mu = \mathbb{E}[Y] = \Pr(Y = 1)$, with $0 < \mu < 1$.

$$\log p_B(y; \mu) = \underbrace{\log \frac{\mu}{1-\mu}}_{\eta} y + \log(1 - \mu).$$

Natural parameter: $\eta := \log \frac{\mu}{1-\mu}$ (log-odds), with $\mu = \sigma(\eta)$.

Step 4: name the coefficient of y in the exponent: the natural parameter η.

Bernoulli: the same three pieces

Recall the binary mean: $\mu = \mathbb{E}[Y] = \Pr(Y = 1)$, with $0 < \mu < 1$.

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$$p_B(y;\eta) = \underbrace{1}_{\text{data only}} \exp\left[\underbrace{\eta y}_{\text{parameter} \times \text{data}} - \underbrace{\log(1+e^{\eta})}_{\text{parameter only}}\right].$$

Step 5: since $1 - \mu = 1/(1+e^{\eta})$, $\log(1-\mu) = -\log(1+e^{\eta})$.

Exponentiating gives the same three pieces as Gaussian.

Bernoulli: the same three pieces

Recall the binary mean: $\mu = \mathbb{E}[Y] = \Pr(Y = 1)$, with $0 < \mu < 1$.

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Back to logistic regression: $\mu = \sigma(z)$, so $\eta = \log \frac{\mu}{1-\mu} = z$.

The predicted mean is $\mu = \sigma(\eta)$; the model sets its natural parameter to the score z.

The shared structure is an exponential-family form

The coefficient of y is the natural parameter η; here both models use $\eta = z$.

$$p_G(y;\eta) = \underbrace{\frac{e^{-y^2/2}}{\sqrt{2\pi}}}_{h_G(y)} \exp\left[\eta y - \underbrace{\frac{\eta^2}{2}}_{A_G(\eta)}\right],$$

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$$p_B(y;\eta) = \underbrace{1}_{h_B(y)} \exp\left[\eta y - \underbrace{\log(1+e^{\eta})}_{A_B(\eta)}\right].$$

Same template: $p(y;\eta) = h(y)\exp[\eta y - A(\eta)]$.

Both examples use the statistic y; next, allow a fixed function $\phi(y)$ in its place. Later, GLMs connect the distribution parameter back to $\mathbf{x}$.

Keep three quantities separate

For one distribution, $\mu = \mathbb{E}[Y]$; for prediction, $\mu(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}]$.

Score z

$z = \mathbf{x}^\top \mathbf{w}$

Produced from $\mathbf{x}$ and $\mathbf{w}$.

Parameter η

$\eta\, \phi(y)$

Multiplies the statistic.

Mean μ

$\mu = \mathbb{E}[Y]$

The prediction target.

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The prediction target.

For these models, $\eta = z$; Gaussian ($\sigma^2 = 1$) uses $\mu = \eta$; Bernoulli uses $\mu = \sigma(\eta)$.

Keep three quantities separate

For one distribution, $\mu = \mathbb{E}[Y]$; for prediction, $\mu(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}]$.

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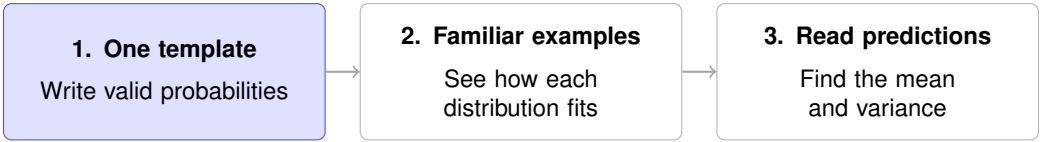
$p(y; \eta)$ is the whole density/probability function. μ is its mean.
For binary labels, μ is also the class-1 probability (the earlier p_n for sample n).

Table of Contents

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- Exponential family
- From Parameter Estimation to GLMs
- Optional reading: extensions and proofs

Where are we? One template for familiar distributions



- **Starting point:** Gaussian and Bernoulli already have the same exponential structure.
- **First:** name the parts and separate what stays fixed from the parameter we change.
- **Then:** see why the normalizing term makes this a valid probability model.

Next question: what does each part of the shared formula do?

Exponential family — four pieces of one distribution

$$p(y;\eta) = \underbrace{h(y)}_{\text{base measure}} \exp\left[\underbrace{\eta\phi(y)}_{\text{parameter–data interaction}} - \underbrace{A(\eta)}_{\text{log-normalizer}}\right].$$

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$\phi(y) \in \mathbb{R}$: fixed functions of y.

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$h(y) \geq 0$: depends on y, not on η.

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$A(\eta) \in \mathbb{R}$: makes total probability one.

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$h(y) \geq 0$: depends on y, not on η.

Log-partition function
$A(\eta) \in \mathbb{R}$: makes total probability one.

Fix the support, h, and ϕ; vary η.
For all three classroom examples, $\phi(y) = y$. Vector extensions are in the optional reading.

Why does the log-partition function normalize the distribution?

The exponential gives nonnegative *weights*; they need not sum to one.

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For Bernoulli, replace the integral by a sum over $y \in \{0, 1\}$:

$$\underbrace{e^{\eta \cdot 0}}_{\text{weight for } y=0} + \underbrace{e^{\eta \cdot 1}}_{\text{weight for } y=1} = \underbrace{1 + e^{\eta}}_{Z(\eta)}.$$

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Divide each weight by Z: $p(0; \eta) = \frac{1}{1+e^{\eta}}$, $p(1; \eta) = \frac{e^{\eta}}{1+e^{\eta}}$; their sum is 1.

Why does the log-partition function normalize the distribution?

The exponential gives nonnegative *weights*; they need not sum to one.

1. Total weight: $Z(\eta) := \int h(y) e^{\eta \phi(y)} \, dy,$

2. Divide by the total: $p(y; \eta) = \dfrac{h(y) e^{\eta \phi(y)}}{Z(\eta)},$

3. Write the division in the exponent: $A(\eta) := \log Z(\eta), \quad e^{-A(\eta)} = \dfrac{1}{Z(\eta)}.$

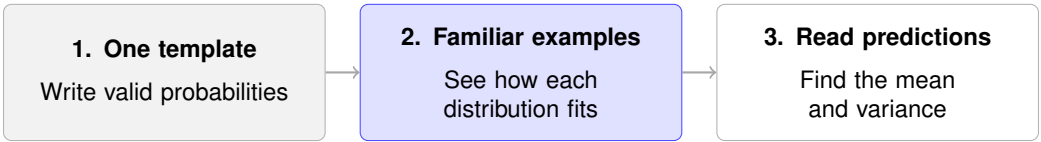
For Bernoulli, replace the integral by a sum over $y \in \{0, 1\}$:

$$\underbrace{e^{\eta \cdot 0}}_{\text{weight for } y=0} + \underbrace{e^{\eta \cdot 1}}_{\text{weight for } y=1} = \underbrace{1 + e^{\eta}}_{Z(\eta)}.$$

Divide each weight by Z: $p(0; \eta) = \frac{1}{1+e^{\eta}}$, $p(1; \eta) = \frac{e^{\eta}}{1+e^{\eta}}$; their sum is 1.

Take-away: $A = \log Z$ turns the total weight into the subtraction needed in the exponent.

Where are we? From a template to familiar distributions

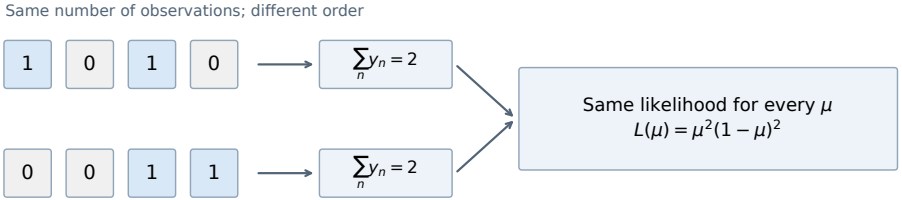


- **So far:** we found the shared template and made its total probability one.
- **Now:** understand the data summary it retains, then identify the same pieces in Bernoulli, Gaussian, and Poisson.

Next question: what information from the observations is needed to fit the distribution?

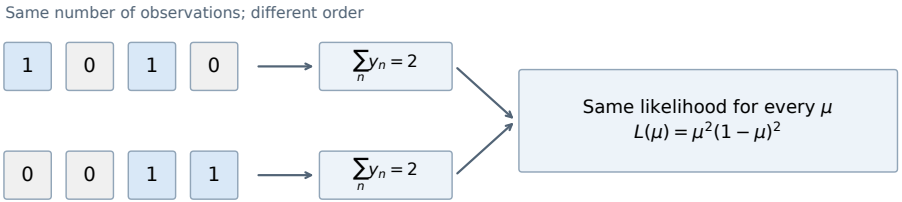
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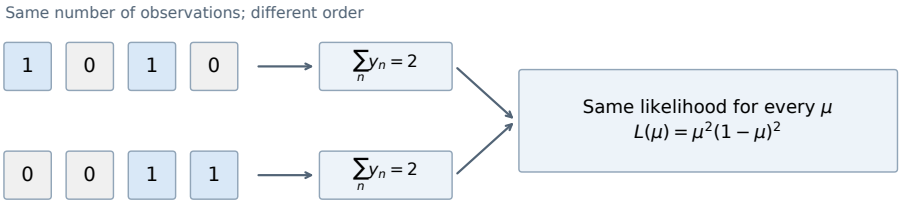


For fixed N, count the ones; the number of zeros is $N - \sum_n y_n$.

$$L(\mu) = \mu^{\sum_n y_n}(1-\mu)^{N-\sum_n y_n}.$$

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For fixed N, count the ones; the number of zeros is $N - \sum_n y_n$.

$$L(\mu) = \mu^{\sum_n y_n}(1-\mu)^{N-\sum_n y_n}.$$

Sufficient: the count contains all the information in these data needed to evaluate the likelihood for μ. The order adds nothing.

Bernoulli: read off the four pieces

We already derived this form from $\mu = \Pr(Y = 1)$:

$$p(y; \eta) = \exp\left[\eta y - \log(1 + e^{\eta})\right], \qquad y \in \{0, 1\}.$$

$$h(y) = 1, \qquad \phi(y) = y,$$

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Same Bernoulli distribution, two parameterizations:
μ is the probability/mean; η is the log-odds. Varying one varies the other.

Gaussian: keep the variance fixed at one

For $Y \sim \mathcal{N}(\mu, 1)$, use the same template with $\phi(y) = y$:

$$p(y;\eta) = \underbrace{\frac{e^{-y^2/2}}{\sqrt{2\pi}}}_{h(y)} \exp\left[\eta y - \underbrace{\frac{\eta^2}{2}}_{A(\eta)}\right].$$

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Bernoulli: $\mu = \sigma(\eta)$. Gaussian (variance 1): $\mu = \eta$.
Same template; a different relation between parameter and mean.

Poisson distributions belong to the exponential family

Recall: A distribution belongs to the exponential family if it can be written in the form

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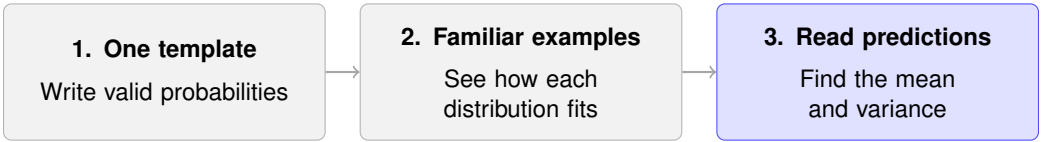
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Where are we? From distributions to predictions



- **So far:** Bernoulli, Gaussian, and Poisson all fit the same template.
- **What changes:** each has its own relation between the natural parameter and the mean.

Next question: can one rule recover the mean and variance in all three?
We return to the log-partition function A.

One function gives us the mean and variance

For our three examples, $\phi(y) = y$. At interior parameters:

$$\underbrace{A'(\eta) = \mathbb{E}[Y] = \mu,}_{\text{first derivative: mean}} \qquad \underbrace{A''(\eta) = \mathrm{Var}(Y)}_{\text{second derivative: variance}} \quad .$$

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For Bernoulli, $A(\eta) = \log(1 + e^{\eta})$:

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Same scalar rule, three distributions

- Bernoulli distribution:

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- Gaussian with variance 1:

$$A(\eta) = \eta^2/2, \qquad A'(\eta) = \eta = \mu, \tag{12}$$

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- Poisson: $A(\eta) = e^{\eta}$, so $A'(\eta) = A''(\eta) = e^{\eta} = \mu$. The mean and variance are equal in this model.

Table of Contents

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- Motivation
- Exponential family
- From Parameter Estimation to GLMs
- Optional reading: extensions and proofs

From distribution properties to learning parameters

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- Then return to prediction: let the parameter depend on the input $\mathbf{x}$. This is the step from fitting a distribution to a GLM.

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$\Longrightarrow$ The cost function J is a convex function in η since $A(\eta)$ is convex.

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At $\bar{y} = 0$ or 1, there is no finite natural-parameter MLE.

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A single fitted distribution has one mean. For prediction, the mean must vary with $\mathbf{x}$.

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- A GLM combines an exponential-family output distribution with a model for its mean.

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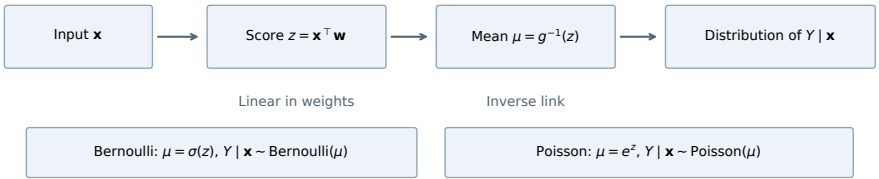
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Set $\eta = \mathbf{x}^\top \mathbf{w}$: then $\mu(\mathbf{x}) = e^{\mathbf{x}^\top \mathbf{w}} > 0$.
Next, formalize this construction using a link function.

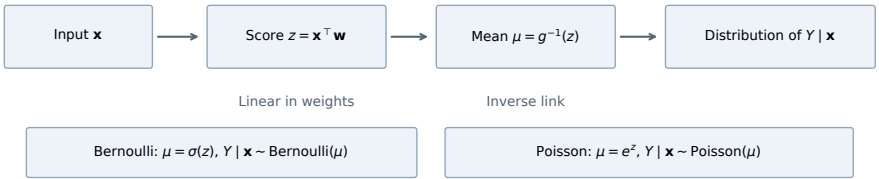
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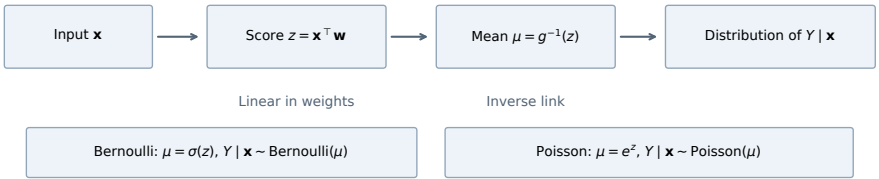
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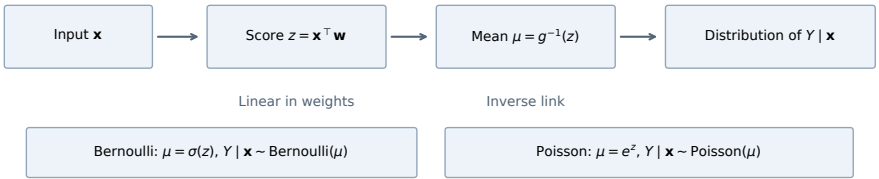
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Quick check: a zero score for a count model

A canonical Poisson GLM has score $z = \mathbf{x}^\top \mathbf{w} = 0$. What is the predicted mean count?

A. 0.

B. 1.

C. $1/2$.

Think / **vote:** choose an answer and give one short reason.

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Answer: B.

The Poisson inverse link is $\mu = e^z$, so $\mu = e^0 = 1$.
The score is the log-mean; it is not the mean count itself.

Before fitting: which parts of the model do we choose?

- Two choice points in the specification of a GLM:
  1. The choice of the exponential family distribution
     $\Longrightarrow$ Generally constrained by the nature of the data Y
  2. The choice of the inverse link g^{-1}
     $\Longrightarrow$ Real degree of freedom!
     $\Longrightarrow$ Canonical response: $g^{-1}(z) = A'(z)$ when $\phi(y) = y$; it sets the natural parameter equal to the linear predictor.

- If we decide to use the canonical inverse link, the choice of the exponential family density completely determines the GLM:

$$p(y|\mathbf{x}; \mathbf{w}) = h(y) \exp\left(\eta\phi(y) - A(\eta)\right) \qquad \text{for} \quad \eta = \mathbf{x}^\top \mathbf{w} \tag{15}$$

Negative log-likelihood estimation

Learn $\mathbf{w}$ by the same likelihood principle.

For a canonical scalar model, $\eta_n = z_n = \mathbf{x}_n^\top \mathbf{w}$, $\phi(y) = y$; $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_N)^\top$.

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Same learning pattern: predicted mean minus observed output, times the input.

Some examples

- Gaussian distribution

Least Squares

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  The fitting objective gains a penalty; choose its strength with validation data.

Table of Contents

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2 Exponential Families and Generalized Linear Models
- Motivation
- Exponential family
- From Parameter Estimation to GLMs
- Optional reading: extensions and proofs

Optional reading: which parameters are allowed?

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We will exclude such parameters by only looking at the set of parameters

$$\text{Allowed parameters: } \left\{\eta : 0 < \int h(y)\exp\left[\eta\phi(y)\right] dy < \infty\right\} \tag{17}$$

Optional reading: Gaussian with two unknown parameters

Two statistics need vectors: $\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) = \eta_1 y + \eta_2 y^2$. The form is

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Recover the original parameters: $\eta_1 = \frac{\mu}{\sigma^2}, \eta_2 = -\frac{1}{2\sigma^2} \iff \mu = -\frac{\eta_1}{2\eta_2}, \sigma^2 = -\frac{1}{2\eta_2}$.

Optional reading: mean-statistic identity (vector form)

$$\nabla_{\boldsymbol{\eta}} A(\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}} \left[\ln \left(\int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) dy \right) \right] \tag{22}$$

$$= \nabla_{\boldsymbol{\eta}} \left[\left(\int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) dy \right) \right] \cdot \left(\int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) dy \right)^{-1} \tag{23}$$

$$= \nabla_{\boldsymbol{\eta}} \left[\left(\int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) dy \right) \right] \cdot \exp\left(-A(\boldsymbol{\eta}) \right) \tag{24}$$

$$= \int \nabla_{\boldsymbol{\eta}} \left[h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) \right] dy \cdot \exp\left(-A(\boldsymbol{\eta}) \right) \tag{25}$$

$$= \int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) \right) \boldsymbol{\phi}(y) dy \cdot \exp\left(-A(\boldsymbol{\eta}) \right) \tag{26}$$

$$= \int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y) - A(\boldsymbol{\eta}) \right) \boldsymbol{\phi}(y) dy \tag{27}$$

$$= \int p(y; \boldsymbol{\eta}) \boldsymbol{\phi}(y) dy = \mathbb{E}_{\boldsymbol{\eta}}[\boldsymbol{\phi}(Y)] \tag{28}$$

For two Gaussian statistics, $\mathbb{E}[\boldsymbol{\phi}(Y)] = (\mu, \mu^2 + \sigma^2)^\top$.

Optional reading: covariance and convexity

From $\log p(y;\boldsymbol{\eta}) = \log h(y) + \boldsymbol{\eta}^\top \boldsymbol{\phi}(y) - A(\boldsymbol{\eta})$,

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This also proves convexity. Zero curvature means some statistic combination is constant; convexity alone does not guarantee a unique or finite MLE.

Optional proof: convexity from normalization

Covariance proved convexity. Here is a direct proof from normalization.

- For $0 < t < 1$ and parameters $\boldsymbol{\eta}_1, \boldsymbol{\eta}_2$, we define $\boldsymbol{\eta} = t\boldsymbol{\eta}_1 + (1-t)\boldsymbol{\eta}_2$. We want to show

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$$A(\boldsymbol{\eta}) \leq tA(\boldsymbol{\eta}_1) + (1-t)A(\boldsymbol{\eta}_2) \tag{29}$$

- We have first

$$\exp A(\boldsymbol{\eta}) = \int h(y) \exp\left(\boldsymbol{\eta}^\top \boldsymbol{\phi}(y)\right) dy \tag{30}$$

$$= \int h(y) \exp\left(\left(t\boldsymbol{\eta}_1 + (1-t)\boldsymbol{\eta}_2\right)^\top \boldsymbol{\phi}(y)\right) dy \tag{31}$$

$$= \int \underbrace{\left[h(y)^t \exp\left(t\boldsymbol{\eta}_1^\top \boldsymbol{\phi}(y)\right)\right]}_{u(y)} \cdot \underbrace{\left[h(y)^{1-t} \exp\left((1-t)\boldsymbol{\eta}_2^\top \boldsymbol{\phi}(y)\right)\right]}_{v(y)} dy \tag{32}$$

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$$\tag{34}$$

Optional proof: convexity from normalization

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Optional proof: apply Hoelder's inequality

For $0 < t < 1$, set $r = 1/t$ and $s = 1/(1-t)$; then $1/r + 1/s = 1$.

The endpoints $t = 0, 1$ are equalities. Taking logs proves convexity.

Optional proof: apply Hoelder's inequality

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$$e^{A(t\boldsymbol{\eta}_1 + (1-t)\boldsymbol{\eta}_2)} \le e^{tA(\boldsymbol{\eta}_1) + (1-t)A(\boldsymbol{\eta}_2)}.$$

The endpoints $t = 0, 1$ are equalities. Taking logs proves convexity.

Optional proof: convexity conclusion

We have

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$$\leq \left(\int h(y) \exp\left(\boldsymbol{\eta}_1^\top \boldsymbol{\phi}(y)\right) dy\right)^t \left(\int h(y) \exp\left(\boldsymbol{\eta}_2^\top \boldsymbol{\phi}(y)\right) dy\right)^{1-t} \tag{38}$$

$$= \exp\left(tA(\boldsymbol{\eta}_1)\right) \exp\left((1-t)A(\boldsymbol{\eta}_2)\right) \tag{39}$$

Taking logs gives $A(t\boldsymbol{\eta}_1 + (1-t)\boldsymbol{\eta}_2) \leq tA(\boldsymbol{\eta}_1) + (1-t)A(\boldsymbol{\eta}_2)$.